



BORN IN **FIRE** BUILT TO LAST

Flanges Division

At **Pragyan Steel & Engineering Company** we take pride in being one of the leading manufacturers of high-quality stainless steel and nickel alloy flanges. With over **three decades of experience**, our flanges are engineered to meet the highest international standards, ensuring exceptional strength, dimensional accuracy, and leak-proof performance in critical service conditions.

Manufacturing Capabilities

- **Size Range:** ½" to 36"
- **Pressure Class:** 150# to 2500#
- **Manufacturing Type:** Forged, Machined, and CNC Finished
- **Material Grades:** Stainless Steel, Carbon Steel, Alloy Steel, Duplex, Super Duplex, Inconel, Monel, Hastelloy, Titanium
- **Standards:** ASTM / ASME / DIN / EN / BS / MSS-SP / ISO

Types of Flanges Manufactured

- Weld Neck Raised Face (**WNRF**)
- Weld Neck Flat Face (**WNFF**)
- Weld Neck Serrated Face (**WNSF**)
- Weld Neck Small Tongue (**WNST**)
- Weld Neck Small Groove (**WNSG**)
- Blind Raised Face (**BLRF**)
- Blind Serrated Face (**BLSG**)
- Blind Large Tongue (**BLLT**)
- Blind Large Groove (**BLLG**)
- Weld Neck Large Tongue (**WNLG**)
- Weld Neck Large Groove (**WNLG**)
- Slip-On, Socket Weld, Lap Joint, Threaded & Orifice Flanges

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Production Process

Our flange manufacturing facility is equipped with **modern CNC, VMC, and turning centers** to ensure precise machining and perfect surface finishing. Every flange is **forged and heat-treated** to achieve the desired mechanical strength and metallurgical properties.

Key Machinery Includes:

- JV55 VMC
- LX20 L3 (2 Units)
- SINEWY 3050 CNC
- Band Saw Machines
- Hydraulic Presses
- Oil & Electric Fired Furnaces
- EOT Cranes & Handling Equipment

Testing & Quality Control

All flanges undergo **comprehensive inspection and testing procedures** to ensure conformance with dimensional and metallurgical requirements.

Testing is performed in accordance with **ASTM, ASME, and PED** standards.

Our Commitment

Every flange manufactured by **Pragyan Steel & Engineering Company** is a blend of **engineering precision and metallurgical excellence**.

We aim to deliver products that **stand the test of time**, ensuring **leak-proof connections, dimensional integrity, and reliable performance** in the most demanding environments.

SEAMLESS CONNECTIONS. INFINITE STRENGTH.

Fittings Division

At **Pragyan Steel & Engineering Company** we manufacture a comprehensive range of **Butt-Weld and Forged Fittings** designed to deliver superior performance in critical process applications. Backed by **three decades of metallurgical expertise** and advanced forming technology, our fittings are known for **dimensional accuracy, corrosion resistance, and long service life** — ensuring leak-proof, high-integrity connections across diverse industries.

Manufacturing Capabilities

- **Size Range:** ½" to 16"
- **Type:** Seamless & Welded Butt-Weld / Forged
- **Material Grades:** Stainless Steel, Alloy Steel, Duplex, Super Duplex, Inconel, Monel, Hastelloy, Titanium, Brass, Cupro Nickel, Incoloy
- **Pressure Ratings:** 3000#, 6000#, 9000#
- **Standards:** ASTM / ASME / MSS-SP / DIN / EN / BS / ISO

Products Manufactured

Butt-Weld Fittings:

- Elbows (45°, 90°, 180° — SR & LR)
- Tees (Equal & Reducing)
- Reducers (Concentric & Eccentric)
- Caps, Stub Ends, Crosses

Forged Fittings:

- Socket Weld & Threaded Elbows, Tees, Couplings, Unions, Plugs, Bushings, Nipples

Each fitting is **formed, machined, and heat-treated** to meet the highest mechanical and metallurgical standards.

Key Machinery Includes:

- Hot Elbow Forming Machine · Cold Elbow Forming Machine
- Cold Forming Hydraulic Press (for Tee, Reducer & Cap)
- Hydraulic Presses · Band Saw Machines
- Electric & Oil-Fired Furnaces · EOT Cranes & Chain Pulley Blocks
- Radial Drill Machines · Lathe Machines



Testing & Quality Control

Every fitting undergoes multi-stage inspection

to ensure conformance with international codes and client specifications.

Testing is carried out by qualified QA/QC personnel as per ISO 9001:2015 and PED requirements.

Tests Conducted:

- Visual & Dimensional Inspection
- Hydrostatic & Pneumatic Testing
- Dye Penetrant & Magnetic Particle Examination
- Mechanical Testing (Tensile, Hardness, Flattening)
- Micro & Macro Examination
- Intergranular Corrosion (IGC) Test
- PMI (Positive Material Identification)

Our Commitment

At **Pragyan Steel & Engineering Company** we believe **quality is engineered** — not inspected.

Every fitting that leaves our facility is a reflection of our promise:

Precision. Strength. Reliability.

*Because when it comes to performance, **Pragyan delivers beyond expectations.***

Because perfection lies where joints don't exist.

STRENGTH THAT FLOWS THROUGH EVERY LINE

Technical Overview

We manufacture Seamless Pipes in Stainless Steel, Nickel Alloys, and Special Metals designed for high-pressure, high-temperature, and corrosive environments. Each pipe undergoes stringent dimensional and metallurgical checks to ensure leak-proof integrity, uniform wall thickness, and high mechanical strength.

Manufacturing Range
Type Outside Diameter Wall Thickness Length
Seamless Pipes 6 mm to 168 mm (¼" to 6") 0.5 mm to 10 mm Up to 6 meters

Welded Pipes ½" NB to 24" NBs per ASTM A312 / A358 Up to 12 meters

Manufacturing Process

- Raw Material: Billets
- Process Route (Seamless): Piercing Rolling
Pilgering Solution Annealing Pickling
Straightening Polishing Inspection Packing

Grades Manufactured

- Austenitic Stainless Steel: SS304 / 304L / 316 / 316L / 321 / 347
- Nickel Alloys: Monel 400, Inconel 600 / 625, Hastelloy C22 / C276, Incoloy 800 / 825
- Duplex & Super Duplex: UNS S31803 / S32750
- Others: Titanium, Cupro Nickel, Brass

Technical Standards

- ASTM: A312 / A213 / A269 / A511 / A790
- ASME: SA312 / SA213 / SA269 / SA511 / SA790
- DIN / EN: EN 10216 / EN 10217
- NACE: MR0175 / MR0103 compliance available

Testing & Inspection

- Hydrostatic Test (100% pipes)
- Eddy Current Test
- Ultrasonic / Pneumatic Leak Test
- PMI (Positive Material Identification)
- Intergranular Corrosion Test (IGC)
- Dimensional & Visual Inspection as per MSS SP-55

Surface Finish

- Bright Annealed (BA)
- Pickled & Passivated
- Polished (Ra Value Upto 0.2)
- Mirror Finish (for instrumentation use)

Applications

- Chemical & Petrochemical Plants
- Nuclear & Defence Installations
- Heat Exchangers & Condensers
- Fertilizer & Refineries
- Power Generation & Process Industries

Seamless performance for every critical application.

PRECISION IN EVERY INCH

Technical Overview

Tubes Division

At **Proficient Tubes Pvt. Ltd.**, a manufacturing unit of **Pragyan Steel & Engineering Company** we specialize in producing **high-performance seamless and welded tubes** designed to meet the most demanding industrial applications. Our manufacturing processes are engineered for **precision, consistency, and reliability**, ensuring that every tube we deliver performs flawlessly under critical conditions.

Manufacturing Capabilities

- **Size Range:** 6 mm OD to 219 mm OD
- **Wall Thickness:** 0.5 mm to 12 mm
- **Length:** Up to 12 meters
- **Manufacturing Type:** Seamless & Welded
- **Grades:** Stainless Steel, Inconel, Monel, Hastelloy, Titanium, Cupro Nickel, and other Nickel Alloys
- **Standards:** ASTM / ASME / DIN / EN / NACE / ISO

Production Process

Our tube manufacturing line is equipped with world-class machinery and process control systems for achieving dimensional accuracy and superior surface finish.

Key Machinery Includes:

- Pilger Mill
- Draw Bench
- Piercing Machine
- Rolling Mill
- Pickling & Passivation Line
- Electric Furnace
- Straightening Machine
- Polishing Machine

Each tube undergoes stringent heat treatment, surface finishing, and inspection protocols to ensure **uniform microstructure, enhanced**

Testing & Quality Control

Quality is integrated at every stage of production. Our in-house testing facilities comply with international standards and customer-specific requirements.

Tests Conducted:

- Hydrostatic & Pneumatic Testing
- Eddy Current Testing
- Ultrasonic Examination
- Visual & Dimensional Inspection
- Mechanical Testing (Tensile, Hardness, Flattening, Flaring)
- Intergranular Corrosion (IGC) Test
- Micro & Macro Examination

All testing is carried out under the supervision of qualified QA personnel as per **ISO 9001:2015** and **PED** requirements.

Applications

Our tubes are widely used across:

- Nuclear & Defence
- Petrochemical & Refinery
- Heat Exchangers & Condensers
- Instrumentation & Control Systems
- Power Generation

Engineered to withstand, designed to deliver.